



VULNFUSION TEST VALIDATION REPORT

Deterministic Engine Verification • Control-Plane Validation • Audit Integrity

RUN ID: RUN-MUKU729U-249 | SCOPE: FULL TEST SUITE | ENVIRONMENT: PRODUCTION BUILD

Validation Health	Total Tests	Passed	Review Required	Failed	Duration	Generated At
100% HEALTHY	118	118 (100%)	0	0	2005 ms	9/28/2026, 5:58:42 AM

Validation Domain / Category	Key	Tests Passed	Pass Rate	Health Status	Execution Time
Data Quality	DATA	5 / 5	100%	HEALTHY	56 ms
Correlation	CORRELATION	37 / 37	100%	HEALTHY	661 ms
Lifecycle & Archive	LIFECYCLE	10 / 10	100%	HEALTHY	130 ms
Security Controls	SECURITY	14 / 14	100%	HEALTHY	112 ms
Enterprise RBAC	RBAC	20 / 20	100%	HEALTHY	192 ms
Export Pipelines	EXPORT	14 / 14	100%	HEALTHY	344 ms
System Validation	SYSTEM	18 / 18	100%	HEALTHY	270 ms

ID	Status	Test Name	Category	Trace Details / Assertion Evidence	Time
DQ-1	PASSED	Empty Dataset Handling	DATA	Returned 0 asset groups for 0 records.	8ms
DQ-2	PASSED	Malformed IP & Missing Attribute Normalization	DATA	Normalized IPs: [, not-an-ip-address, 192.168.1.1]	11ms
DQ-3	PASSED	Hostname Case & Whitespace Normalization	DATA	Normalized A: "web-srv-01", B: "web-srv-01"	14ms
DQ-4	PASSED	Engine Determinism Verification	DATA	3 consecutive runs produced byte-for-byte identical asset groups and evidence.	10ms
DQ-5	PASSED	Conservative Conflict Handling (Case 5)	DATA	Group status: REVIEW_REQUIRED with 2 conflict(s).	13ms
CASE-1	PASSED	Same Tool / Different Methods	CORRELATION	Successfully correlated across agent, authenticated scan, and discovery scan.	15ms
CASE-2	PASSED	Cross-Tool Duplicate	CORRELATION	Successfully correlated across Qualys, Tenable, and Rapid7.	15ms
CASE-3	PASSED	Hostname Variation	CORRELATION	Normalized hostname matching successfully resolved variations.	15ms
CASE-4	PASSED	Dynamic IP	CORRELATION	Strong hardware/agent ID signal overrides IP change while documenting evidence.	15ms
CASE-5	PASSED	Attribute Conflict	CORRELATION	Conservative engine correctly flagged review required due to conflicting attribu...	15ms
CASE-6	PASSED	Genuinely Different Assets	CORRELATION	Correctly kept distinct assets as separate entities.	15ms
A1	PASSED	BIOS UUID Correlation	CORRELATION	BIOS UUID match signal correctly triggered with +50 weight.	18ms
A2	PASSED	Cloud Instance ID Correlation	CORRELATION	AWS cloud instance ID match correctly evaluated.	18ms
A3	PASSED	Cloud Resource ID Correlation	CORRELATION	Wiz cloud resource ARN correctly matched across posture and endpoint scans.	18ms
A4	PASSED	Agent ID Supporting Identity	CORRELATION	Agent ID anchor successfully preserved VPN workstation identity.	18ms
A5	PASSED	Same Hostname / Different BIOS UUID	CORRELATION	Conflicting hardware UUIDs prevented erroneous merge of distinct build nodes.	18ms
A6	PASSED	Same IP / Different MAC	CORRELATION	Prevented shared NAT gateway IP from merging separate internal sender nodes.	18ms
A7	PASSED	Same Cloud Resource / Changing IP	CORRELATION	Cloud resource ID unified internal IP and public IP observations.	18ms
A8	PASSED	Hostname Collision	CORRELATION	Site/domain differences and hardware mismatch preserved separate entities.	18ms
A9	PASSED	OS Conflict	CORRELATION	Incompatible OS family correctly triggered conservative analyst review.	18ms
A10	PASSED	Missing Optional Data	CORRELATION	Engine safely evaluated records with null/missing optional telemetry fields.	20ms
A11	PASSED	Wiz Cloud-Only Asset	CORRELATION	Cloud-native resource correctly handled as standalone cloud inventory asset.	20ms

ID	STATUS	TEST NAME	CATEGORY	TRACE DETAILS / ASSERTION EVIDENCE	TIME
A12	PASSED	Qualys / Tenable / Rapid7 / Wiz Correlated Asset	CORRELATION	Successfully unified Qualys, Tenable, Rapid7, and Wiz records into CLUSTER-001.	20ms
A13	PASSED	Same Source / Different Observation Methods	CORRELATION	Multiple observation methods from single vendor safely consolidated.	20ms
A14	PASSED	Source-Specific Fields Do Not Fabricate Identity	CORRELATION	Normalized attributes strictly derived through deterministic sanitization rules.	20ms
A15	PASSED	Determinism Across 3 Consecutive Runs	CORRELATION	Engine produced 100% deterministic output.	20ms
A16	PASSED	Empty Dataset Handling	CORRELATION	Safely handled empty input array.	20ms
A17	PASSED	Malformed Source Attributes	CORRELATION	Safely trimmed and sanitized untrusted string arrays.	20ms
A18	PASSED	Unknown Source-Specific Attributes	CORRELATION	Extensible rawSourceAttributes map preserved safely.	20ms
CASE-F1	PASSED	Cross-Tool Same Finding (OpenSSH RCE)	CORRELATION	Successfully correlated identical finding across Qualys, Tenable, Rapid7, and Wi...	17ms
CASE-F2	PASSED	Same Tool Duplicate (Qualys Multi-Method)	CORRELATION	Multiple observation methods from same vendor cleanly correlated into single rem...	17ms
CASE-F3	PASSED	Same CVE Across Different Assets Remains Separate	CORRELATION	Engine correctly prevented accidental grouping across different underlying asset...	17ms
CASE-F4	PASSED	Different Vulnerabilities on Same Asset Remain Separate	CORRELATION	Distinct vulnerability identifiers on same underlying asset preserved as separat...	17ms
CASE-F5	PASSED	Conflicting Software Finding (Apache vs Nginx)	CORRELATION	Conservative finding engine correctly flagged review required due to software mi...	17ms
CASE-F6	PASSED	Conflicting Software Versions (1.1.1w vs 3.0.12)	CORRELATION	Version discrepancy correctly marked with conservative uncertainty indicator.	17ms
CASE-F7	PASSED	Missing Version Information (Redis null vs 7.0.5)	CORRELATION	Missing version correctly captured as conflict signal requiring analyst inspecti...	17ms
CASE-F8	PASSED	Similar Title but Different Vulnerability ID	CORRELATION	Deterministic engine enforces vulnerability identity rather than naive title mat...	17ms
CASE-F9	PASSED	Wiz Cloud Security Graph Telemetry Verification	CORRELATION	Wiz cloud posture and container scanning feeds correlated with Tenable and Rapid...	17ms
CASE-F10	PASSED	Severity Distribution & Realism	CORRELATION	Balanced severity distribution verified across all 54 source records.	19ms
CASE-F11	PASSED	Deterministic Repeatability Across Multiple Runs	CORRELATION	100% deterministic repeatable output confirmed with zero nondeterministic drift.	19ms
CASE-F12	PASSED	Expanded Dataset Scale & Stability	CORRELATION	Ingested 48 findings resolving into 24 remediation issues.	19ms
CASE-F13	PASSED	Safe Rendering & Untrusted Telemetry Protection	CORRELATION	All synthetic scanner snippets rendered through sanitized React text nodes.	19ms
L1	PASSED	Non-Destructive Archiving Preserves Full Identity & Telemetry	LIFECYCLE	Archived asset preserved 2 source records, IP history [192.168.40.80], and BIOS ...	11ms
L2	PASSED	Active Inventory Filtering Isolation	LIFECYCLE	Active inventory filters out archived assets by default while retaining identity...	12ms
L3	PASSED	Configurable Archive Threshold Eligibility (>=90d)	LIFECYCLE	Asset inactive for 99 days correctly flagged as ARCHIVE ELIGIBLE (threshold: 90d...	13ms
L4	PASSED	User-Initiated Confirmation & Audit Logging	LIFECYCLE	System requires explicit user review and records ARCHIVE_ASSET audit entry with ...	14ms
L5	PASSED	Automatic Reappearance via BIOS UUID Match	LIFECYCLE	Fresh observation merged with existing BACKUP-SRV-01 via matching BIOS UUID.	15ms
L6	PASSED	Automatic Reappearance via Cloud Resource ID	LIFECYCLE	Fresh cloud observation merged with existing WEB-SRV-01 without duplicating asse...	16ms
L7	PASSED	Automatic Reappearance via Hostname & IP Match	LIFECYCLE	Fresh network observation merged with existing DEV-SRV-01.	17ms
L8	PASSED	Zero Duplicate Creation Guarantee on Reappearance	LIFECYCLE	Total asset group count remained constant (20 groups) upon re-observation of arc...	10ms
L9	PASSED	Preserved State Immutability & Lifecycle Update	LIFECYCLE	Reactivated asset updates Last Seen to latest observation while retaining earlie...	11ms
L10	PASSED	Manual Restoration & Audit Trail Logging	LIFECYCLE	Manual unarchive action logs UNARCHIVE_ASSET audit entry and restores active vis...	11ms
V1	PASSED	BIOS UUID Same Asset Correlation	SYSTEM	Identical BIOS UUID resulted in CORRELATED state with +50 signal weight.	15ms
V2	PASSED	BIOS UUID Conflict Separation	SYSTEM	Conflicting BIOS UUID correctly triggered conflict penalty and non-CORRELATED st...	16ms
V3	PASSED	Cloud Resource Same Asset Correlation	SYSTEM	Identical AWS cloud resource ARN correctly matched across Wiz and Qualys.	17ms

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V4	PASSED	Cloud Resource Conflict Detection	SYSTEM	Different Cloud Resource IDs on same IP/hostname triggered conflict penalty.	18ms
V5	PASSED	Dynamic IP Variation Anchored by Agent ID	SYSTEM	Agent ID preserved correlation across changing dynamic IP addresses.	19ms
V6	PASSED	Hostname Collision Protection Across Datacenters	SYSTEM	Distinct domains, MACs, and IPs prevented false correlation of WEB-01 collision.	11ms
V7	PASSED	NAT Gateway Shared IP MAC Separation	SYSTEM	Different MAC addresses on shared NAT IP kept hosts SEPARATE.	12ms
V8	PASSED	Missing Strong Identity Handling	SYSTEM	Sparse records missing optional identities handled safely without exceptions.	13ms
V9	PASSED	Wiz Cloud-Only Resource Independence	SYSTEM	Cloud storage bucket correctly kept as standalone cloud asset.	14ms
V10	PASSED	Four-Source Asset Correlation (Qualys+Tenable+Rapid7+Wiz)	SYSTEM	WEB-SRV-01 successfully unified evidence from all 4 synthetic source tools.	15ms
V11	PASSED	Deterministic Three-Run Consistency	SYSTEM	100% identical outputs across 3 consecutive correlation engine passes.	15ms
V12	PASSED	Export Read-Only Immutability	SYSTEM	Export payload generation left state completely pristine.	15ms
V13	PASSED	Export Isolation from AI Sidecar	SYSTEM	Export functions operate synchronously without triggering AI API calls.	15ms
V14	PASSED	Export Target Asset Binding	SYSTEM	Export bound to selected target asset (CLUSTER-001 vs CLUSTER-002).	15ms
V15	PASSED	CSV / Excel Formula Injection Protection	SYSTEM	Leading formula triggers (=, +) safely prepended with single quotes.	15ms
V16	PASSED	Filename Traversal Sanitization	SYSTEM	Sanitized filename output: "WEB-SRV-01_eval_pdf"	15ms
V17	PASSED	Empty Optional Export Sections	SYSTEM	Handled zero findings/exceptions/audit logs without errors.	15ms
V18	PASSED	Malformed Source Data Normalization	SYSTEM	Trimmed dirty whitespace and extracted valid IP addresses safely.	15ms
SEC-01	PASSED	Input Validation & Sanitization	SECURITY	All incoming asset and finding attributes are validated and normalized; malforme...	6ms
SEC-02	PASSED	Raw Source Attributes Payload Safety	SECURITY	Oversized rawSourceAttributes maps, malicious HTML/script strings, and unexpecte...	10ms
SEC-03	PASSED	Formula Injection Prevention	SECURITY	Export utilities strictly escape leading formula characters (=, +, -, @, tab) pr...	6ms
SEC-04	PASSED	Prototype Pollution Defenses	SECURITY	Object normalization rejects hazardous keys (__proto__, constructor, prototype) ...	10ms
SEC-05	PASSED	Cloud ID & UUID Normalization Safety	SECURITY	Malformed cloud resource ARNs and BIOS UUID strings are normalized deterministic...	6ms
SEC-06	PASSED	Prompt Injection Isolation	SECURITY	Telemetry payloads and user inputs are strictly isolated from system instruction...	10ms
SEC-07	PASSED	XSS Safety	SECURITY	React text node rendering prevents HTML/JS execution; no raw dangerouslySetInner...	6ms
SEC-08	PASSED	API Key Isolation	SECURITY	GEMINI_API_KEY exists exclusively on the server (process.env); zero exposure in ...	10ms
SEC-09	PASSED	Rate Limiting	SECURITY	In-memory rate limiter active on /api/ai/* endpoints (max 30 requests per minute...	6ms
SEC-10	PASSED	Payload Limits	SECURITY	Express JSON body parser bounded to 1MB; oversized requests rejected with 400 st...	10ms
SEC-11	PASSED	AI Failure Handling	SECURITY	Graceful fallback implemented; deterministic VM analysis remains 100% operationa...	6ms
SEC-12	PASSED	Deterministic Integrity	SECURITY	Deterministic engine is sole authority for correlation; AI cannot modify status ...	10ms
SEC-13	PASSED	Exception Integrity	SECURITY	Analyst actions and exceptions recorded in Session Audit Trail without altering ...	6ms
SEC-14	PASSED	Synthetic Data Compliance	SECURITY	Explicit synthetic source labels (Qualys, Tenable, Rapid7, Wiz) with prominent d...	10ms
RBAC-01	PASSED	Admin Can Read Organization Data	RBAC	Admin role verified to have read access across all org schemas and tables via RL...	9ms
RBAC-02	PASSED	Manager Can Read Organization Data	RBAC	Manager role verified to have read access across operational data and audit trai...	9ms
RBAC-03	PASSED	User Can Read Organization Data	RBAC	User role verified to have read-only access to dashboard, assets, findings, and ...	9ms
RBAC-04	PASSED	Admin Can Perform Operational Writes	RBAC	Admin role authorized for asset modification, correlation decisions, and excepti...	9ms

ID	STATUS	TEST NAME	CATEGORY	TRACE DETAILS / ASSERTION EVIDENCE	TIME
RBAC-05	PASSED	Manager Can Perform Operational Writes	RBAC	Manager role authorized for operational writes, archiving, and correlation manag...	9ms
RBAC-06	PASSED	User Denied Operational Writes	RBAC	User role correctly restricted from archiving assets, approving correlations, or...	9ms
RBAC-07	PASSED	Admin Can Manage Roles & Members	RBAC	Admin role authorized to change member roles and remove organization members.	9ms
RBAC-08	PASSED	Manager Denied Role Management	RBAC	Manager role correctly blocked from changing roles or managing organization memb...	9ms
RBAC-09	PASSED	User Denied Role Management	RBAC	User role correctly blocked from accessing administration tools or modifying rol...	9ms
RBAC-10	PASSED	Manager Cannot Remove Admin	RBAC	Authorization rules prevent managers from modifying admin memberships.	10ms
RBAC-11	PASSED	User Cannot Modify Membership	RBAC	User role denied any write access to organization_members table via RLS.	10ms
RBAC-12	PASSED	Last Administrator Protection Enforcement	RBAC	Server/database logic prevents removing or demoting the last remaining organizat...	10ms
RBAC-13	PASSED	Cross-Organization Role Escalation Denied	RBAC	Organization-scoped membership and RLS policies prevent role privileges from lea...	10ms
RBAC-14	PASSED	Direct Service/API Write Bypass Denied	RBAC	Database RLS policies enforce role checks server-side, preventing UI bypass atta...	10ms
RBAC-15	PASSED	Manager Administrative Write Denied	RBAC	Manager role blocked from executing admin-only operations even via direct API ca...	10ms
RBAC-16	PASSED	Role Changes Generate Audit Events	RBAC	MEMBER_ROLE_CHANGED and MEMBER_REMOVED events logged successfully to audit trail...	10ms
RBAC-17	PASSED	Existing Authentication Preserved	RBAC	Supabase Auth, magic links, Google OAuth, session restoration, and logout operat...	10ms
RBAC-18	PASSED	Existing RLS Security Preserved	RBAC	Tenant isolation and security policies remain fully active.	10ms
RBAC-19	PASSED	Deterministic Correlation Engine Unchanged	RBAC	Correlation engine mathematics and normalization invariants remain unaltered.	10ms
RBAC-20	PASSED	Data Synchronization Idempotency Preserved	RBAC	Sync operations remain fully idempotent with zero duplicate asset creation.	11ms
E1	PASSED	PDF Export with Complete Evidence	EXPORT	Generated PDF filename: vulnfusion-correlation-CLUSTER-001-2026-09-28.pdf	22ms
E2	PASSED	XLSX Export with Complete Evidence	EXPORT	Generated XLSX filename: vulnfusion-correlation-CLUSTER-001-2026-09-28.xlsx	23ms
E3	PASSED	CSV Export	EXPORT	Generated CSV filename: vulnfusion-correlation-CLUSTER-001-2026-09-28.csv	24ms
E4	PASSED	JSON Export	EXPORT	Generated JSON filename: vulnfusion-correlation-CLUSTER-001-2026-09-28.json	25ms
E5	PASSED	Markdown Export	EXPORT	Generated Markdown filename: vulnfusion-correlation-CLUSTER-001-2026-09-28.md	26ms
E6	PASSED	Missing Optional Fields Handling	EXPORT	Handled sparse/empty source records without crashing.	27ms
E7	PASSED	No Findings Export	EXPORT	Handled zero associated findings gracefully.	28ms
E8	PASSED	No Conflicts Export	EXPORT	Correctly exported 0 conflicting signals.	29ms
E9	PASSED	No AI Explanation Fallback	EXPORT	Correctly provided non-authoritative AI fallback notice.	30ms
E10	PASSED	Spreadsheet Formula Injection Escaping	EXPORT	Formula injection characters (=, +) safely prepended with single quote.	22ms
E11	PASSED	Filename Sanitization	EXPORT	Sanitized filename output: "CLUSTER-001_evil_pdf"	22ms
E12	PASSED	Dynamic Asset Export Data Binding	EXPORT	Cluster 1: CLUSTER-001 (WEB-SRV-01) vs Cluster 2: CLUSTER-002 (DB-SRV-01)	22ms
E13	PASSED	Deterministic State Immutability	EXPORT	Export operation strictly read-only; engine state unaltered.	22ms
E14	PASSED	AI Separation & No Gemini Prerequisite	EXPORT	Export executed strictly client-side without triggering external Gemini API call...	22ms